

CHAPTER – 1

INTRODUCTION TO EMBEDDED SYSTEM

1.1 INTRODUCTION:

An embedded system is a specialized computing system designed to perform a specific task or a set of tasks within a larger system. Unlike general-purpose computers, embedded systems are dedicated to particular functions and are optimized for performance, efficiency, and reliability.

Embedded systems consist of both hardware and software components. The hardware typically includes a microcontroller or microprocessor, memory, input/output interfaces, and other supporting components. The software is programmed to control the system and perform the desired operations.

These systems are widely used in everyday life and can be found in devices such as washing machines, microwave ovens, automobiles, medical equipment, and industrial machines. Each embedded system is designed to operate with minimal human intervention and often works in real-time environments, where quick and accurate responses are essential.

One of the key features of embedded systems is their ability to interact with the physical world through sensors and actuators. Sensors collect data from the environment, and the embedded system processes this data to make decisions and control outputs accordingly. With the advancement of technology, embedded systems have become more powerful and are now integrated with communication technologies like the Internet of Things (IoT). This allows devices to connect, share data, and be monitored remotely, making systems smarter and more efficient.

In applications such as precision agriculture, embedded systems play a crucial role in monitoring environmental conditions like temperature, humidity, and soil moisture. By using microcontrollers such as ESP32, developed by Espressif Systems, these

systems can collect real-time data and transmit it to cloud platforms for analysis and decision-making.

1.2 EMBEDDED SYSTEM:

An embedded system is a specialized computing system designed to perform a specific function or a set of functions within a larger system. It consists of both hardware and software components, where the hardware typically includes a microcontroller or microprocessor, memory, and input/output interfaces, and the software programmed to control the system operations. Unlike general-purpose computers, embedded systems are task-oriented and operate with high efficiency and reliability, often in real-time environments. These systems interact with the physical world through sensors and actuators, enabling them to monitor and control various processes. Embedded systems are widely used in everyday devices such as washing machines, automobiles, medical equipment, and smart agricultural systems, making them an essential part of modern technology.

EXAMPLES OF EMBEDDED SYSTEM:

1. Washing Machine

- Uses an embedded system to control washing cycles, water level, and timing automatically.

2. Microwave Oven

- Controls cooking time, temperature, and power levels using a microcontroller.

3. Engine Control Unit (ECU)

- Manages engine functions like fuel injection and ignition timing in vehicles.

4. Traffic Light Control System

- Automatically controls traffic signals based on timing or sensors.

5. ATM Machine

- Performs banking operations such as cash withdrawal and balance checking.

1.3 CHARACTERISTICS:

1. Dedicated Functionality:

- Embedded systems are designed for a specific purpose and optimized for that single use case, unlike general-purpose computers.

2. Real-Time Performance:

- Many embedded systems must respond to events within tight time constraints to ensure safe and correct operation.

3. Low Power Consumption:

- Embedded devices often run on batteries or have limited power budgets, necessitating efficient hardware and software design.

4. Compact Size:

- Embedded systems are miniaturized for integration into larger products or for use in space-constrained environments.

5. Reliability:

- Since embedded systems often control critical functions, they must operate reliably over long periods without failure.

6. Firmware-Based Operation:

- Programs run from ROM or Flash memory and are not typically changed by end users.

Embedded systems are designed to perform specific tasks rather than be general-purpose computers for multiple tasks. Some also have real-time performance constraints that must be met for reasons such as safety and usability. The software written for embedded systems is often called firmware, and is usually stored in read-only memory or Flash memory chips. It often runs with limited hardware resources: small or no keyboard, screen, and little memory.

1.4 MICRO PROCESS(MP):

A microprocessor is a general-purpose digital computer central processing unit (CPU). It contains an arithmetic and logical unit (ALU), a program counter (PC), a stack pointer (SP), some working registers, a clock timing circuit, and interrupt circuits. The microprocessor requires external memory, I/O ports, and peripherals to function as a complete computing system.

Schematic Arrangement of a Microprocessor Based System

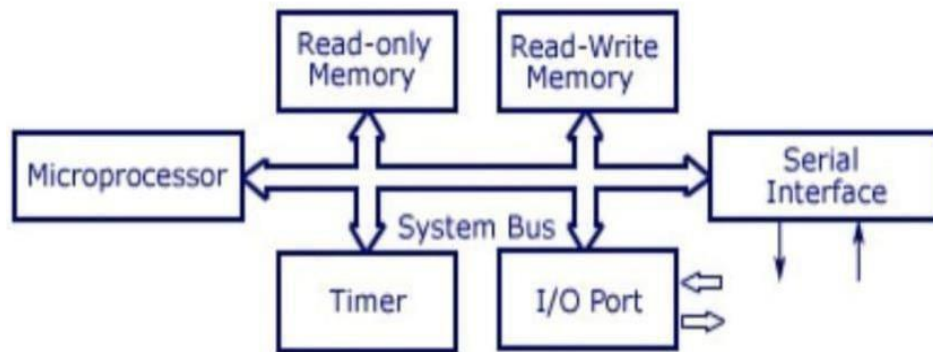


Figure 1.1:Schematic Arrangement of a Microprocess Based System

1.5 MICROCONTROLLER(MC):

A microcontroller incorporates all the features found in a microprocessor CPU: ALU, PC, SP, and registers, plus additional features needed to make a complete computer including ROM, RAM, parallel I/O ports, serial ports, counters, timers, and in some cases, an analog-to-digital converter. The STM32 is a powerful microcontroller based on the ARM Cortex-M core, offering rich peripheral sets and high performance for embedded control applications.

Schematic Internal Architecture of a Microcontroller

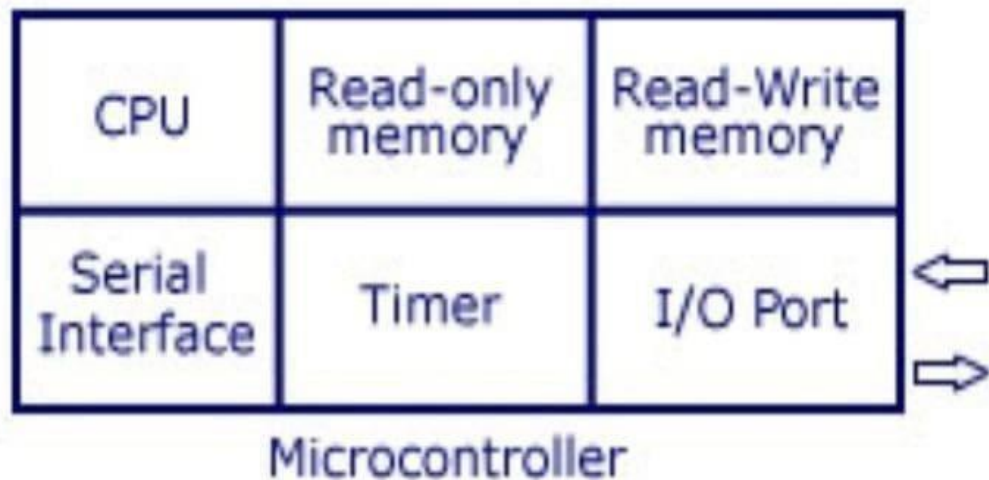


Figure 1.2:schematic Internal Architecture of a Microcontroller

1.6 DIFFERENCE BETWEEN MICROPROCESSOR AND MICROCONTROLLER

The microprocessor must have many additional external parts to be operational as a computer, whereas the microcontroller requires no additional external digital parts. The prime use of microprocessors is to read data, perform extensive calculations, and store or display results. The prime function of a microcontroller is to read data, perform limited calculations, and control its environment based on these results. The microcontroller is a special-purpose digital controller, while the microprocessor is a general-purpose CPU.

Parameter	Microprocessor	Microcontroller
Purpose	General-purpose CPU	Special-purpose controller
Memory	External RAM/ROM required	On-chip ROM, RAM, Flash
I/O Ports	External I/O chips required	Built-in I/O ports
Cost	Higher (external components)	Lower (all-in-one chip)
Power	Higher consumption	Low power consumption
Speed	Very high (multi-GHz)	Moderate (MHz range)
Size	Larger due to external parts	Compact — single chip
Programming	OS-based (Windows, Linux)	Firmware / Embedded C
Examples	Intel Core, AMD Ryzen	STM32, Arduino, PIC

Table 1.1: Difference Between Microprocessor and Microcontroller

1.7 PERIPHERALS:

Embedded Systems communicate with the outside world via peripherals, such as:

- Serial Communication Interfaces (SCI): RS-232, RS-422, RS-485
- Synchronous Serial Communication Interfaces: I2C, JTAG, SPI, SSC
- Universal Serial Bus (USB)
- Networks: Ethernet, Controller Area Network, LAN networks
- Timers: PLL(s), Capture/Compare and Time Processing Units
- Discrete IO: General Purpose Input/Output (GPIO)
- Analog to Digital / Digital to Analog (ADC/DAC)

CHAPTER – 2

OVERVIEW OF THE PROJECT

2.1 INTRODUCTION:

The rapid advancement of technology has led to the development of smart systems that improve efficiency and productivity in various fields, including agriculture. Precision agriculture is an emerging approach that uses modern technologies to monitor and manage environmental conditions for better crop yield and resource optimization.

This project focuses on the development of a smart environmental monitoring system using the ESP32 microcontroller. The system is designed to measure important environmental parameters such as temperature, humidity, and soil moisture in real time. These parameters play a crucial role in determining plant health and agricultural productivity.

The system uses sensors like DHT11/DHT22 for temperature and humidity measurement and a soil moisture sensor to detect the water content in the soil. The collected data is processed by the ESP32 microcontroller, developed by Espressif Systems, and transmitted to IoT platforms through built-in Wi-Fi.

Using cloud platforms such as ThingSpeak and Blynk, the system allows users to monitor environmental conditions remotely via mobile or web applications. This helps in making informed decisions regarding irrigation and crop management.

The proposed system is cost-effective, easy to implement, and suitable for both small-scale and large-scale agricultural applications. It reduces manual effort, improves efficiency, and supports sustainable farming practices.

Background of the Project:

Agriculture is one of the most important sectors for sustaining human life and supporting the economy. Traditional farming methods largely depend on manual monitoring and natural conditions, which can lead to inefficiencies in resource usage and reduced crop productivity. Factors such as temperature, humidity, and soil moisture play a crucial role in plant growth, but maintaining optimal levels of these parameters is often challenging for farmers.

With the advancement of technology, modern agriculture is gradually shifting towards automation and data-driven decision-making. Precision agriculture has emerged as an effective approach that uses advanced .

The processing, and wireless communication. By integrating sensors and communication technologies, farmers can remotely monitor field conditions and take timely actions.

In this context, the development of a smart environmental monitoring system becomes essential. Such a system helps in continuously tracking temperature, humidity, and soil moisture, providing accurate data for better decision-making. It reduces dependency on manual observation and supports efficient resource management.

Motivation:

Agriculture plays a vital role in supporting the growing population, but traditional farming methods often rely on manual monitoring and guesswork. This can lead to inefficient use of resources such as water and energy, as well as reduced crop productivity. Farmers may not always be able to accurately monitor environmental conditions like temperature, humidity, and soil moisture, which are essential for healthy plant growth.

The need for a smart and automated solution to overcome these challenges is the primary motivation behind this project. With the advancement of embedded systems and IoT technologies, it is now possible to continuously monitor environmental parameters in real time. The use of microcontrollers like the ESP32, developed by Espressif Systems, allows efficient data collection and wireless communication.

This project is motivated by the goal of developing a low-cost, easy-to-use, and energy-efficient system that can help farmers make informed decisions. By providing real-time data and remote monitoring capabilities, the system aims to improve water management, reduce manual effort, and enhance crop yield

Problem Statement:

Agriculture heavily depends on environmental factors such as temperature, humidity, and soil moisture, which directly influence crop growth and productivity. In traditional farming methods, these parameters are monitored manually, which is time-consuming, labor-intensive, and often inaccurate. Farmers may not be able to continuously observe field conditions, leading to improper irrigation, water wastage, and reduced crop yield.

Additionally, the lack of real-time monitoring and remote access to environmental data makes it difficult to take timely decisions. Variations in climate conditions and unpredictable weather further increase the challenges faced by farmers. In many cases, overwatering or underwatering occurs due to the absence of proper soil moisture monitoring systems.

There is a need for an efficient, low-cost, and automated solution that can continuously monitor environmental parameters and provide real-time data to users. With the help of embedded systems and IoT technologies, such the ESP32 microcontroller developed by Espressif Systems, it is possible to design a system that can collect, process, and transmit environmental data wirelessly.

Therefore, the main problem addressed in this project is the lack of an efficient and reliable system real-time monitoring of environmental conditions in agriculture. The proposed solution aims to overcome these limitations by providing a smart, automated, and IoT-based monitoring system to improve agricultural productivity and resource management.

Objectives of the Project:

The primary objective of this project is to design and develop a smart environmental monitoring system using the ESP32 microcontroller for precision agriculture. The system is intended to provide an efficient and reliable solution for monitoring important environmental parameters such as temperature, humidity, and soil moisture in real time. These parameters play a crucial role in plant growth, and maintaining their optimal levels is essential for improving crop productivity and

ensuring sustainable agricultural practices.

Another important objective of this project is to implement accurate data acquisition using sensors such as DHT11 or DHT22 for temperature and humidity measurement, and a soil moisture sensor for determining the water content in the soil. These sensors continuously collect data from the environment and send it to the ESP32 microcontroller. The ESP32, developed by Espressif Systems, processes this data efficiently and prepares it for further transmission.

The project also aims to enable wireless communication using the built-in Wi-Fi feature of the ESP32. This allows the collected data to be transmitted to cloud-based IoT platforms such as ThingSpeak and Blynk. Through these platforms, users can monitor environmental conditions remotely using smartphones or computers. This remote monitoring capability reduces the need for constant physical presence in the field and allows farmers to take timely actions based on real-time data.

Another key objective is to improve water management in agriculture. By continuously monitoring soil moisture levels, the system helps in determining the exact water requirements of crops. This prevents both overwatering and underwatering, thereby conserving water resources and improving plant health. Efficient irrigation practices not only save water but also reduce operational costs for farmers.

The project further aims to develop a system that is cost-effective, energy-efficient, and easy to implement. The use of affordable components like ESP32 and basic sensors makes the system accessible to small and medium-scale farmers. Additionally, the low power consumption of the ESP32 ensures that the system can operate for long durations, even in remote areas where power availability may be limited. Another objective is to enhance crop productivity and quality by maintaining optimal environmental conditions. By providing accurate and real-time data, the system helps farmers make informed decisions regarding irrigation, crop management, and environmental control. This leads to better yield and improved agricultural outcomes.

Finally, the project aims to demonstrate the practical application of embedded systems and IoT technologies in agriculture. It provides a foundation for future enhancements such as automation of irrigation systems, integration of advanced

sensors, and implementation of data analytics for predictive farming.

Another objective is to create a scalable system that can be extended with additional features such as wireless communication and IoT integration.

Finally, the project aims to test and validate the system to ensure accurate performance and reliability under different conditions, demonstrating its practical usability.

Scope of the Project:

The scope of this project is focused on the design and development of a smart environmental monitoring system using the ESP32 microcontroller for precision agriculture. The system is intended to monitor important environmental parameters such as temperature, humidity, and soil moisture in real time, which are essential for effective crop management.

The project includes the integration of sensors such as DHT11/DHT22 for measuring temperature and humidity, and a soil moisture sensor for determining the water content in the soil. These sensors collect data from the environment and send it to the ESP32 microcontroller, developed by Espressif Systems, which processes the data and prepares it for transmission.

Another important aspect within the scope is the use of wireless communication through the built-in Wi-Fi of the ESP32. This enables the system to send data to IoT cloud platforms such as ThingSpeak and Blynk. Through these platforms, users can monitor environmental conditions remotely using mobile devices or computers, making the system convenient and user-friendly.

The project mainly focuses on monitoring and data visualization rather than full automation. However, it provides a strong foundation for future expansion, such as automatic irrigation control, integration of additional sensors (like pH, light, or rainfall sensors), and the use of advanced technologies like artificial intelligence for predictive analysis. The system is designed to be cost-effective, energy-efficient, and suitable for small to medium-scale agricultural applications. It can be deployed in farms, greenhouses, and gardens to assist farmers in maintaining optimal conditions for crop growth. However, the scope of this project is limited by factors

such as dependency on internet connectivity for IoT functionality and the use of basic sensors, which may have limited accuracy compared to advanced industrial-grade sensors. Despite these limitations, the project demonstrates a practical and scalable solution for smart farming.

2.2 EXISTING SYSTEM:

Before the adoption of integrated microcontroller platforms like STM32, magnetic field detection and contactless control systems were implemented using conventional and complex methods. Traditional systems relied on mechanical switches, infrared sensors, or bulky electromagnetic sensor arrays.

In many existing obstacle or magnetic detection setups, mechanical relays and simple threshold comparators were used. These systems are prone to mechanical wear, require physical contact, and suffer from limited sensitivity and response speed. IR-based systems are significantly affected by ambient light conditions, limiting their practical utility.

Camera-based vision systems provide detailed spatial information but require high processing power, complex algorithms, and are expensive. They also struggle in poor lighting conditions and consume more energy, making them less suitable for simple, low-cost embedded applications.

Overall, existing systems suffer from issues such as contact-based wear, lower accuracy, higher cost, sensitivity to environmental conditions, limited flexibility, and increased system complexity. These limitations create the need for a more reliable, affordable, and efficient solution like the STM32-based smart contactless control system with Hall effect and touch sensors.

Drawbacks of Existing System

- Contact-based wear and tear — mechanical switches degrade over time.
- Lower accuracy — threshold comparators lack fine-grained measurement.
- Higher cost — camera systems and complex sensor arrays increase budget.
- Limited flexibility — single-mode detection does not adapt to varied use cases.

- Increased system complexity — multiple components with wiring challenges.

2.3 PROPOSED SYSTEM:

The proposed system is a smart environmental monitoring system designed using the ESP32 microcontroller for precision agriculture. The system aims to monitor important environmental parameters such as temperature, humidity, and soil moisture in real time and provide accurate data to users for better decision-making.

The system consists of sensors such as DHT11/DHT22 for measuring temperature and humidity, and a soil moisture sensor for detecting the water content in the soil. These sensors continuously collect data from the environment and send it to the ESP32 microcontroller, developed by Espressif Systems. The ESP32 processes the data and prepares it for transmission.

The processed data is transmitted wirelessly using the built-in Wi-Fi module of the ESP32. The system is connected to IoT cloud platforms such as ThingSpeak and Blynk, where the data is stored and displayed in the form of graphs and real-time values. This enables users to monitor environmental conditions remotely through mobile devices or web applications.

The proposed system reduces the need for manual monitoring and provides continuous and accurate data. It helps farmers optimize irrigation by analyzing soil moisture levels, thereby conserving water and improving crop productivity. The system is cost-effective, energy-efficient, and easy to implement, making it suitable for both small-scale and large-scale agricultural applications.

2.3.1 BLOCK DIAGRAM OF PROPOSED SYSTEM

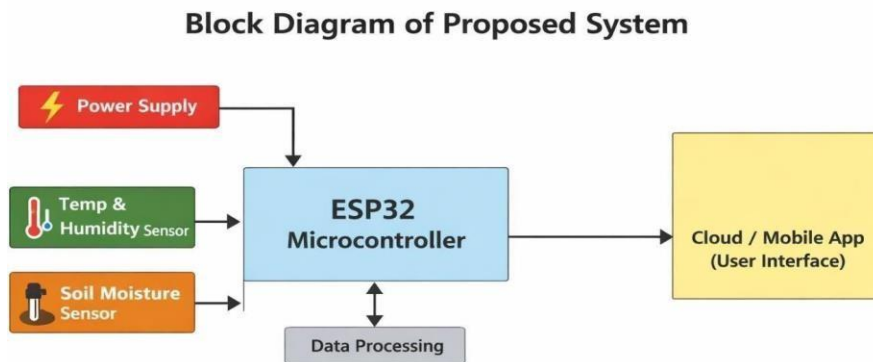


Figure 2.1:Block Diagram of Proposed System

Here's a detailed description and design for the block diagram of the proposed ESP32 -based smart environmental monitoring system for precision agriculture. I'll provide both the conceptual layout and an explanation of each block so it's clear for your report.

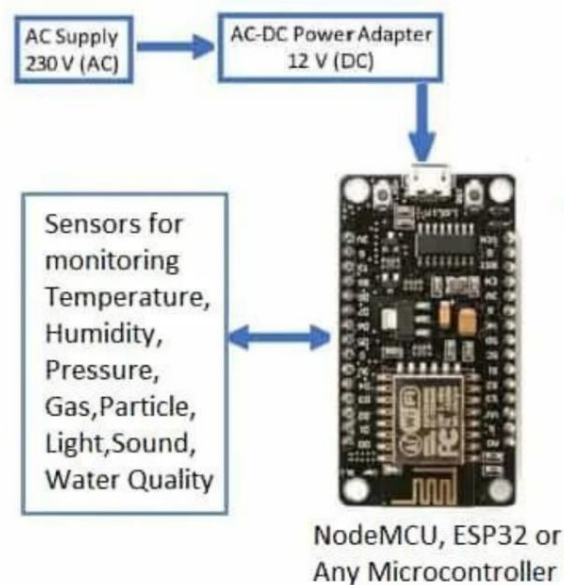


Figure 2.2:Block diagram of the proposed ESP32 Based Smart Environmental Monitoring System for Precision Agriculture

HARDWARE COMPONENTS:

- ESP32 Microcontroller
- DHT11/DHT22 Temperature and Humidity Sensor
- Soil Moisture Sensor
- Power Supply Unit
- Breadboard
- Jumper Wires

SOFTWARE COMPONENTS:

- Arduino IDE
- Embedded C Programming Language
- ESP32 Board Package
- Thing Speak
- Blynk App

CHAPTER – 3

LITERATURE REVIEW

3.1 INTRODUCTION

The literature review provides an overview of existing research and developments related to smart environmental monitoring systems and precision agriculture. With the rapid advancement of embedded systems and Internet of Things (IoT) technologies, several studies have been conducted to improve agricultural productivity through real-time monitoring and automation.

Researchers have explored various methods to monitor environmental parameters such as temperature, humidity, and soil moisture using different types of sensors and microcontrollers. Systems based on microcontrollers like Arduino and ESP32, developed by Espressif Systems, have gained significant attention due to their low cost, ease of use, and built-in communication capabilities.

Many existing systems utilize IoT platforms such as ThingSpeak and Blynk for data storage, visualization, and remote monitoring. These platforms allow users to access real-time data from anywhere, enabling better decision-making in agricultural practices.

Previous research has also highlighted the importance of automated irrigation systems based on soil moisture levels. Such systems help in conserving water and improving crop yield. However, some limitations have been identified, including dependency on internet connectivity, sensor accuracy issues, and system scalability.

This project builds upon the existing research by developing a smart environmental monitoring system using ESP32 that focuses on real-time data acquisition, wireless communication, and user-friendly monitoring. The literature review helps in understanding current technologies, identifying research gaps, and providing a foundation for the proposed system.

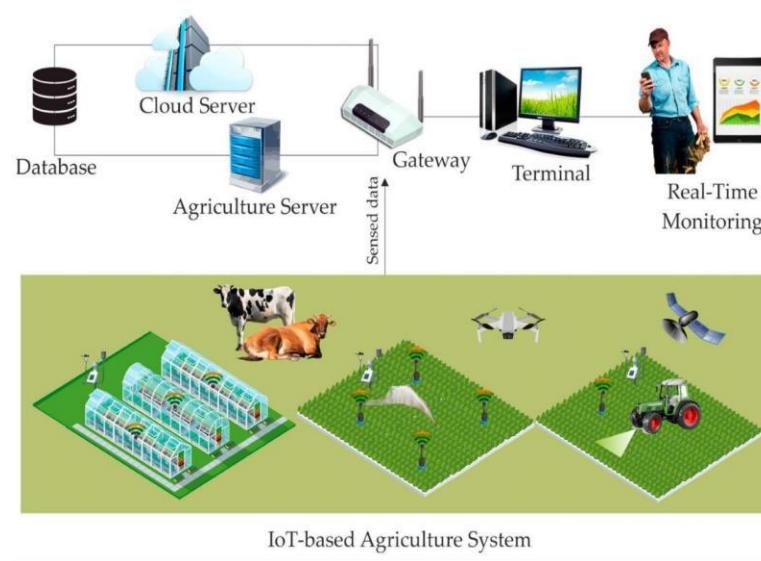


Figure 3.1: IOT Based Agriculture System

3.2 DHT11/DHT22

The DHT11 and DHT22 are digital sensors used to measure temperature and humidity. They are very popular in IoT and embedded systems projects because they are simple to use and provide calibrated digital output.

Working Principle

These sensors combine two main components:

1. Thermistor

- Measures temperature
- Resistance changes with temperature

2. Capacitive Humidity Sensor

- Measures moisture in the air
- Capacitance changes with humidity

3. Built-in Microcontroller

- Converts analog signals into digital data
- data using a single-wire communication protocol

4. Pin Configuration

Both sensors usually have 3 or 4 pins:

- VCC → Power supply (3.3V or 5V)
- DATA → Digital output signal
- GND → Ground
- (Optional) NC → Not connected

How It Works (Step-by-Step)

1. Microcontroller sends a start signal
2. Sensor detects signal and wakes up
3. Sensor measures:
 - Temperature
 - Humidity
4. Data is converted into digital format
5. Sensor sends data back through the DATA pin

Advantages

- Easy to interface with microcontrollers (like ESP32)
- Digital output (no ADC needed)
- Low cost and widely available
- Pre-calibrated sensor
- Good for real-time monitoring

Disadvantages

- Slow response time
- Limited sampling rate

- DHT11 has lower accuracy
- Not ideal for extreme conditions

Applications

- Weather monitoring systems
- Smart agriculture
- Home automation
- Greenhouse monitoring
- IoT environmental sensing

3.3 SOIL MOISTURE:

A soil moisture sensor is a device used to measure the water content in soil. It helps determine whether the soil is dry, moist, or wet, making it essential for smart irrigation and agriculture systems.

Working Principle

There are two main types of soil moisture sensors:

1. Resistive Soil Moisture Sensor

- Uses two metal probes inserted into the soil
- Measures electrical resistance
- Wet soil → Low resistance → High conductivity
- Dry soil → High resistance → Low conductivity

2. Capacitive Soil Moisture Sensor

- Measures capacitance (dielectric constant) of soil
- More accurate and durable
- Not affected by corrosion like resistive sensors

Pin Configuration

Most soil moisture modules have 4 pins:

- VCC → Power supply (3.3V / 5V)

- GND → Ground
- AO (Analog Output) → Gives moisture level (continuous value)
- DO (Digital Output) → Gives HIGH/LOW based on threshold

How It Works (Step-by-Step)

- Sensor probes are inserted into the soil
- Soil moisture affects conductivity/capacitance
- Sensor converts this into an electrical signal
- Output is sent to microcontroller (ESP32)
- ESP32 reads:
- Analog value → exact moisture level
- Digital value → dry/wet status

Advantages

- Helps automate irrigation systems
- Saves water
- Improves crop yield
- Easy to interface with ESP32/Arduino
- Low cost and widely available

Disadvantages

- Resistive sensors corrode over time
- Needs calibration for accurate readings
- Readings vary with soil type
- Sensitive to environmental conditions

Applications

- Smart irrigation systems
- Precision agriculture
- Greenhouse monitoring

- Plant watering automation
- Environmental monitoring

ESP32 Connection (Important for Report)

- VCC → 3.3V
- GND → GND
- AO → GPIO 34 (ADC pin)
- DO → GPIO 2 (optional)

Output Interpretation (Example)

Condition	Analog Value Range
Dry Soil	3000 – 4095
Moist Soil	1500 – 3000
Wet Soil	0 – 1500



Figure 3.2:Block Diagram of Proposed System

3.4 ROLE OF STM32 IN CONTACTLESS APPLICATIONS:

The ESP32 microcontroller, developed by Espressif Systems, plays a significant role in enabling contactless applications due to its built-in Wi-Fi and Bluetooth capabilities. These features allow devices to communicate wirelessly without physical interaction, making ESP32 highly suitable for modern smart systems.

Key Roles of ESP32 in Contactless Applications

1. Wireless Communication

- ESP32 supports Wi-Fi and Bluetooth (BLE)
- Enables communication between devices without physical contact
- Used in applications like contactless monitoring and control

2. IoT Integration

- Connects devices to cloud platforms like ThingSpeak and Blynk
- Allows remote data access and control
- Essential for smart and automated systems

3. Real-Time Data Monitoring

- Collects data from sensors (temperature, humidity, etc.)
- Sends real-time updates to users
- Eliminates the need for manual inspection

4. Touchless Control Systems

- Can be integrated with:
 - Motion sensors
 - IR sensors
 - Mobile apps
- Enables touch-free operations (e.g., automatic doors, smart switches)

5. Low Power Operation

- Suitable for battery-powered contactless devices
- Supports sleep modes for energy efficiency

6. Smart Automation

- Enables automation without human interaction
- Used in:
 - Smart homes
 - Healthcare systems
 - Agriculture monitoring

Applications of ESP32 in Contactless Systems

1. Contactless temperature monitoring
2. Smart attendance systems
3. Automatic sanitizers
4. Touchless door systems
5. Smart irrigation systems

CHAPTER – 4

TECHNOLOGY & TOOLS

4.1 TECHNOLOGIES USED

1.DHT11/22

- The proposed system is developed using a combination of embedded systems, sensor technologies, and Internet of Things (IoT) platforms.
- These technologies work together to enable real-time environmental monitoring and remote data access.

2.Capacitive Touch Sensing Technology

- The TTP223 capacitive touch sensor detects changes in capacitance due to finger proximity or touch.
- It provides a contactless input mechanism enabling gesture-based control.
- The sensor output is a digital signal compatible with ESP32 GPIO pins.

3.Embedded Systems Technology

- The ESP32 microcontroller acts as an embedded system that receives data from sensors, processes it, and controls outputs like LED indicators and buzzers.
- Embedded systems are ideal for real-time control in compact hardware.

4.Microcontroller Technology (ARM Cortex-M)

- The ESP32's ARM Cortex-M core is the brain of the system.
- It reads sensor data, performs calculations, makes decisions, and drives output devices.

5.I2C / SPI Communication Technology

- The OLED display communicates with the ESP32 using the I2C protocol.
- I2C allows multiple devices to communicate using just two wires (SDA and SCL).

6.ADC (Analog-to-Digital Conversion)

- The ESP32's built-in ADC converts the analog output of the Hall effect sensor to digital values.
- This enables precise measurement of magnetic field intensity.

7.Interrupt-Driven Processing

- The ESP32 uses hardware interrupts to respond immediately to sensor events.
- This ensures real-time detection without busy-waiting loops.

8.Obstacle/Field Detection Technology

- The software compares the measured Hall voltage with a predefined threshold.
- If a magnetic field is detected, it triggers the buzzer and/or LED alert.

4.2 TOOLS REQUIRED

1. Hardware Tools

The following hardware tools are required for the implementation of the proposed system:

- ESP32 Microcontroller
- DHT11/DHT22 Temperature and Humidity Sensor
- Soil Moisture Sensor
- Breadboard
- Jumper Wires
- Power Supply (Battery/USB Adapter)
- USB Cable (for programming and power)

2. Software Tools

The following software tools are used for the development and implementation of the proposed system:

- Arduino IDE
- Embedded C Programming Language
- ESP32 Board Package
- ThingSpeak
- Blynk

CHAPTER – 5

HARDWARE IMPLEMENTATION

5.1 HARDWARE COMPONENTS ARE:

- ESP32 Microcontroller
- DHT11/DHT22 Temperature and Humidity Sensor
- Soil Moisture Sensor
- Breadboard
- Jumper Wires
- Power Supply (Battery/USB Adapter)
- USB Cable

5.2 ESP32 MICROCONTROLLER

The ESP32 is a powerful and low-cost microcontroller used in embedded systems and IoT applications. It is developed by Espressif Systems and is widely used due to its high performance and built-in wireless communication features such as Wi-Fi and Bluetooth.

ESP32 MICROCONTROLLER – SPECIFICATION AND VALUE

Specification	Value
Microcontroller	Tensilica Xtensa Dual-Core
Operating Voltage	3.3 V
Input Voltage	5 V (via USB)
Clock Frequency	Up to 240 MHz
Flash Memory	4 MB (typical)
SRAM	~520 KB
Wi-Fi	802.11 b/g/n

Specification	Value
Bluetooth	Bluetooth v4.2 (BLE + Classic)
GPIO Pins	Up to 34
ADC Channels	12-bit ADC
DAC Channels	2 (8-bit)
Communication Protocols	UART, SPI, I2C, CAN
Power Consumption	Low (with sleep modes)
Operating Temperature	-40°C to +125°C

Table 5.1:ESP32 Microcontroller Parameters



Figure5.1: ESP32 Microcontroller

ADVANTAGES OF ESP32 MICROCONTROLLER

1. Built-in Wi-Fi and Bluetooth

- Enables wireless communication without extra modules

2. High Processing Speed

- Dual-core processor up to 240 MHz

3. Low Power Consumption

- Supports deep sleep and power-saving modes

4. Cost-Effective

- Affordable for IoT and embedded projects

5. Multiple GPIO Pins

- Supports interfacing with many sensors and devices

6. Supports Multiple Protocols

- UART, SPI, I2C, CAN communication

7. Compact Size

- Suitable for small and portable systems

8. Integrated ADC and DAC

- Supports both analog input and output

DISADVANTAGES OF ESP32 MICROCONTROLLER

1. Complex for Beginners

- More advanced compared to basic microcontrollers like Arduino UNO

2. Higher Power Usage than Simple MCUs

- Consumes more power than low-end controllers (e.g., ATmega328)

3. Limited Analog Accuracy

- ADC readings may not be very precise

4. Heat Generation

- Can generate heat during continuous operation

5. Wi-Fi Dependency Issues

- Performance depends on network availability

5.3 DHT11/DHT22 TEMPERATURE AND HUMIDITY SENSOR

The DHT11 and DHT22 are digital sensors used to measure temperature and humidity in environmental monitoring systems. These sensors are widely used in embedded and IoT applications due to their simplicity, reliability, and low cost.

Features

- Digital output signal
- Measures both temperature and humidity
- Easy to interface with microcontrollers
- Low power consumption
- Cost-effective

Specifications and Values

Parameter	DHT11	DHT22
Temperature Range	0°C to 50°C	-40°C to 80°C
Humidity Range	20% to 80%	0% to 100%
Accuracy	±2°C, ±5% RH	±0.5°C, ±2% RH
Output	Digital	Digital
Operating Voltage	3.3V to 5V	3.3V to 6V

Table 5.2: DHT11/DHT22 Parameters

Working Principle

The sensor contains a humidity sensing component and a thermistor. It measures environmental conditions and sends the data as a digital signal to the microcontroller, which then processes and displays the values.

Applications

- Weather monitoring systems
- Smart agriculture
- Home automation
- Environmental monitoring

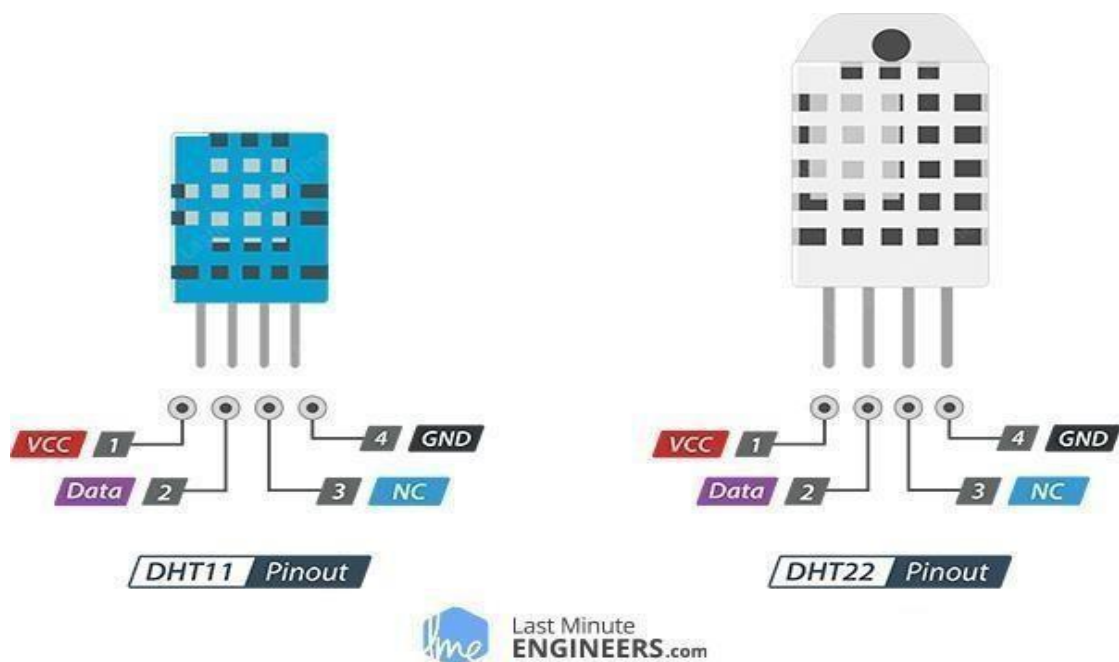


Fig: 5.2 DHT11/DHT22 Temperature and Humidity Sensors

ADVANTAGES OF ESP32

1. Built-in Wi-Fi and Bluetooth

- No need for external communication modules

2. High Speed Performance

- Dual-core processor up to 240 MHz

3.Low Power Consumption

- Supports sleep modes for energy saving

4.Cost-Effective

- Affordable for IoT and embedded projects

5.Multiple GPIO Pins

- Easy interfacing with sensors and devices

6.Supports Multiple Communication Protocols

- UART, SPI, I2C, CAN

7.Compact and Lightweight

- Suitable for portable applications

8.Integrated ADC and DAC

- Supports analog and digital signals

DISADVANTAGES OF ESP32

1.Complex for Beginners

- Requires more understanding than basic microcontrollers

2.Higher Power Usage (Compared to Basic MCUs)

- Consumes more power than simple controllers like Arduino UNO

3.ADC Accuracy is Limited

- Not suitable for high-precision analog readings

4.Heat Generation

- Can heat up during continuous operation

5.Wi-Fi Dependency

- Requires stable internet for IoT applications

5.4 SOIL MOISTURE SENSOR

A soil moisture sensor is used to measure the water content present in the soil. It plays a vital role in smart agriculture by helping to determine when irrigation is needed. The sensor provides real-time data about soil conditions, which helps in efficient water management.

Features

1. Measures soil moisture level
2. Easy to interface with microcontrollers
3. Provides analog and digital output
4. Low power consumption
5. Cost-effective

Specifications and Values

Parameter	Value
Operating Voltage	3.3V to 5V
Output Type	Analog and Digital
Working Current	~10–20 mA
Detection Range	Depends on soil condition
Interface	GPIO (Analog/Digital pins)

Table 5.3: Soil Moisture Specification and Values

The sensor works based on the electrical conductivity of the soil. When the soil contains more water, it conducts electricity better, resulting in higher moisture readings. Dry soil has lower conductivity, indicating low moisture levels. The sensor sends this data to the microcontroller for processing.

Applications

1. Smart irrigation systems
2. Precision agriculture
3. Garden monitoring
4. Greenhouse systems

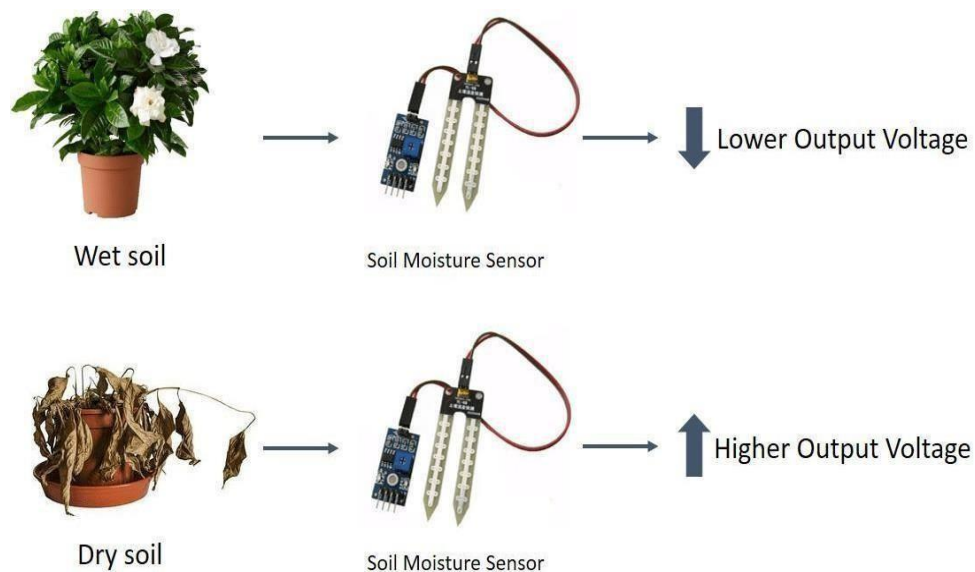


Figure 5.3: Soil Moisture Advantages of Soil Moisture Sensor

ADVANTAGES

1. Helps in efficient water management
2. Provides real-time soil moisture data
3. Easy to interface with microcontrollers (like ESP32)
4. Low cost and easily available
5. Improves crop productivity
6. Supports automation of irrigation systems
7. Low power consumption

DISADVANTAGES OF SOIL MOISTURE SENSOR

1. Accuracy may vary depending on soil type
2. Prone to corrosion (especially resistive type)
3. Requires calibration for better results
4. Affected by environmental conditions
5. Limited lifespan due to continuous exposure to moisture

5.5 OLED DISPLAY

An OLED display uses Organic Light Emitting Diodes to produce light without a backlight, making it thinner, more energy-efficient, and capable of showing high contrast. In electronics projects, small OLED modules like the SSD1306 (128x64) are commonly used. These displays communicate with microcontrollers using I2C or SPI protocols. OLED screens are ideal for showing text, sensor values, and simple graphics. Their compact size and clear visibility make them perfect for real-time data display in embedded projects.

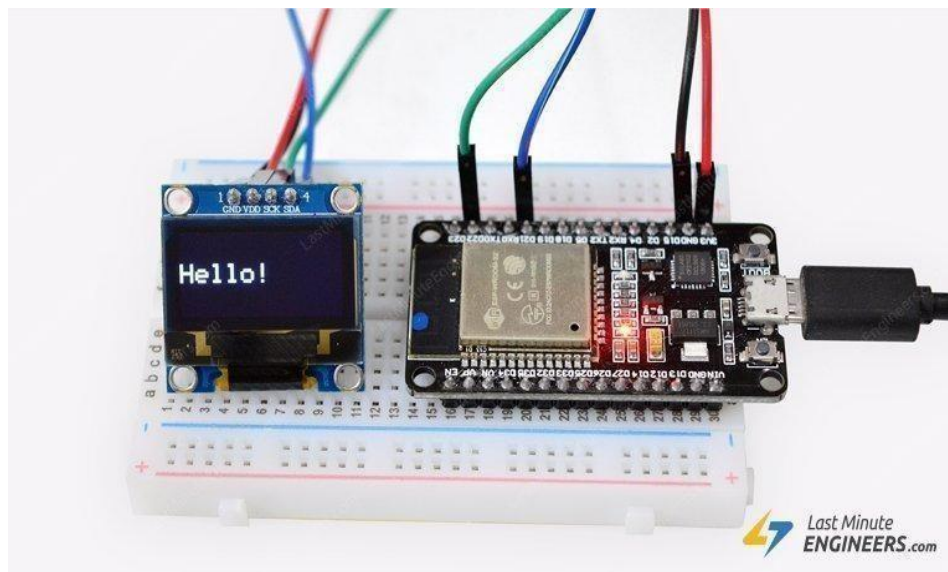


Figure5.4 : Breadboard

Specification	Value
Display Technology	OLED (monochrome)
Resolution	128 × 64 pixels
Controller IC	SSD1306
Communication	I2C (0x3C / 0x3D) or SPI
Operating Voltage	3.3V – 5V
Display Size	0.96 inch diagonal
Contrast Ratio	Very high (no backlight needed)
Power Consumption	~20 mA(typical)
Operating Temperature	-40°C to +85°C

Table 5.4:OLED Specifications and Values



Figure 5.5: OLED Display

5.6 BUZZER

A buzzer is an electronic audio signaling device used to produce sound alerts. In this project, the buzzer provides audible feedback when a magnetic field is detected or a touch gesture is registered. Active buzzers produce sound when power is applied, while passive buzzers require a PWM signal to generate tones. The buzzer operates at low voltage and is easy to interface with ESP32 GPIO pins. They are small, inexpensive, and consume very little power.

Parameter	Active Buzzer	Passive Buzzer
Activation	Direct DC power	PWM signal needed
Frequency	Fixed (built-in)	Variable (user-defined)
Programming	Simple GPIO HIGH/LOW	Requires timer/PWM setup
Sound Output	Single tone	Multiple tones / melodies
Cost	Slightly higher	Lower
Use in Project	Primary alert	Optional enhanced feedback

Table 5.5: Buzzer Parameters



Figure 5.6: Buzzer

5.7 JUMPER WIRES

Jumper wires are flexible electrical wires used to make temporary connections in electronic circuits. They are commonly used with breadboards and development boards. They come in three types: male-to-male, male-to-female, and female-to-female. The colored insulation helps identify signals like power, ground, and data lines. They are essential in prototyping and educational electronics projects.



Figure5.7:Jumper Wires

5.8 POWER SUPPLY

A power supply provides the electrical energy required by the circuit components. The ESP32 operates at 3.3V logic, powered via a micro-USB or external 5V adapter with an onboard 3.3V regulator. Stable power supply is critical for accurate sensor readings and reliable system operation. The Dht11/22 sensor and Soil Moisture sensor are compatible with 3.3V to 5V supply.

5.9 USB CABLE

A USB cable provides both power and data communication between the ESP32 and a computer. It is used to upload programs via the ST-LINK programmer and debug the system. The USB cable ensures a stable power source during development and testing. It is widely available, cost-effective, and essential for ESP32 programming and monitoring.



Figure 5.8:USB Cable

CHAPTER – 6

SOFTWARE IMPLEMENTATION

6.1 INTRODUCTION TO ESP32 ARDUINO IDE

Software implementation is one of the most important parts of the proposed energy monitoring system. It controls the complete working of the project by collecting sensor data, processing the readings, displaying the output, and sending the measured values to the IoT platform. In this project, the software is developed using the Arduino IDE to program the ESP32 microcontroller. The program is designed to read voltage and current values from the ZMPT101B voltage sensor and ACS712 current sensor, calculate electrical parameters such as voltage, current, power, and energy, display them on the 16x2 LCD with I2C, and upload the readings to the Blynk IoT application.

Key Features of STM32CubeIDE

1. Easy Programming Environment

- Uses Arduino IDE, which is simple and beginner-friendly

2. Support for Embedded C/C++

- Programming using Arduino-based syntax

3. Built-in Library Support

- Provides libraries for Wi-Fi, Bluetooth, sensors, and displays

4. Wi-Fi Connectivity

- Easy integration of internet-based applications

5. Bluetooth (BLE & Classic)

- Supports wireless communication projects

6. Multiple GPIO Control

- Easy handling of input/output pins

7. Analog and Digital Support

- Supports ADC, DAC, PWM operations

8. Serial Communication

- Supports UART, SPI, and I2C communication

9. Cross-Platform Compatibility

- Works on Windows, Linux, and macOS

10. Open Source Platform

- Free to use with large community support

6.2 SOFTWARE TOOLS USED:

The software for the proposed system is developed using the Arduino Integrated Development Environment (IDE). It is a simple and widely used platform for writing, compiling, and uploading code into microcontrollers such as the ESP32

Arduino IDE is selected for this project because it supports:

- ESP32 programming
- LCD interfacing
- sensor reading Wi-Fi communication
- IoT integration

It also provides a Serial Monitor, which is useful for checking sensor values and debugging the program during testing.

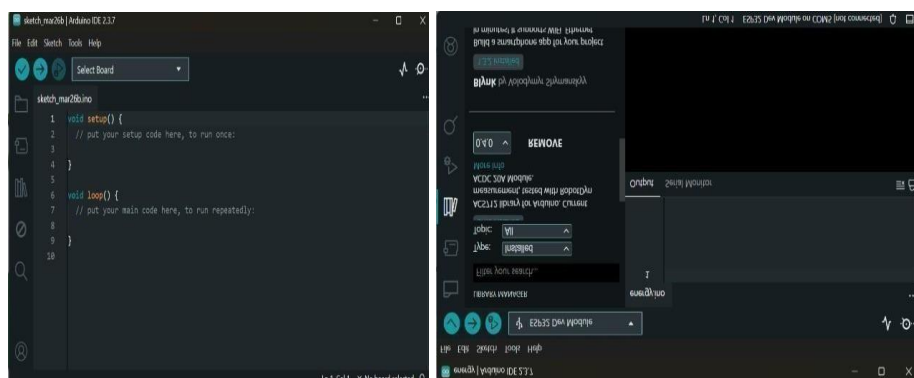
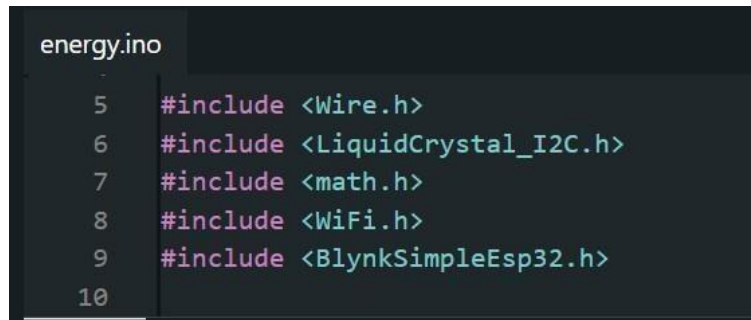


Figure 6.1: Arduino IDE Environment

6.3 LIBRARIES AND INITIAL SETUP

To implement the software successfully, different libraries are included in the program. These libraries help the ESP32 communicate with the LCD display, Wi-Fi network, and Blynk platform.



```
energy.ino
5  #include <Wire.h>
6  #include <LiquidCrystal_I2C.h>
7  #include <math.h>
8  #include <WiFi.h>
9  #include <BlynkSimpleEsp32.h>
10
```

Figure 6.2: Required libraries

The LCD is initialized with I2C address 0x27, and the analog input pins of the ESP32 are assigned for voltage and current sensor readings. for the voltage sensor and GPIO35 is used for the current sensor. In the program, GPIO34 is used. The Wi-Fi credentials and Blynk authentication details are also included in the software so that the ESP32 can connect to the internet and send the measured data to the IoT platform.

FINAL CODE

```
#include "DHT.h"

// --- Configuration ---
#define DHTPIN 2      // Digital pin for DHT sensor
#define DHTTYPE DHT11 // Use DHT11 or DHT22
#define SOILPIN A0    // Analog pin for Soil Moisture sensor
```



```
DHT dht(DHTPIN, DHTTYPE);

void setup() {
  Serial.begin(9600);
  Serial.println("Sensor Test Start");
  dht.begin();
}

void loop() {
  // Wait a few seconds between measurements
  delay(2000);

  // --- Read DHT Sensor ---
  float h = dht.readHumidity();
  float t = dht.readTemperature(); // Celsius

  // --- Read Soil Moisture ---
  int soilMoisture = analogRead(SOILPIN);

  // Map value: usually 1023 (dry) to 300 (wet)
  int soilPercent = map(soilMoisture, 1023, 300, 0, 100);

  // Check if reads failed
  if (isnan(h) || isnan(t)) {
    Serial.println("Failed to read from DHT sensor!");
    return;
  }

  // --- Print Results ---
  Serial.print("Humidity: ");
  Serial.print(h);
```

```
Serial.print("% Temperature: ");  
Serial.print(t);  
Serial.print(" °C ");  
  
Serial.print("Soil Moisture: ");  
Serial.print(soilPercent);  
Serial.println("%");  
}
```

OUTPUT

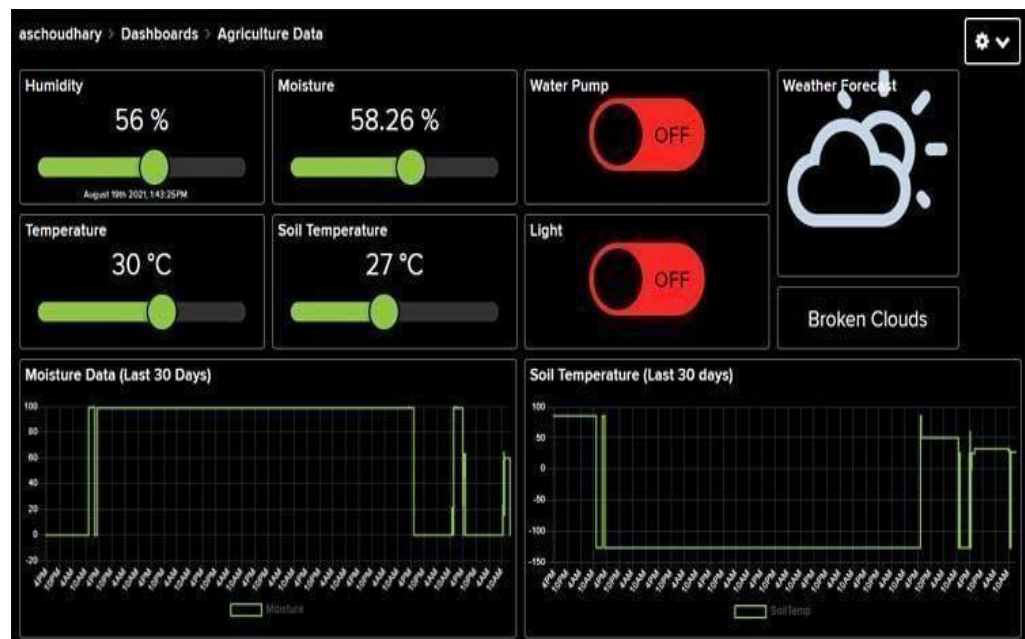


Figure 6.3:DHT11/DHT22

The developed smart environmental monitoring system using the ESP32 microcontroller was successfully implemented and tested in real-time conditions. The system effectively monitored temperature, humidity, and soil moisture levels, demonstrating reliable performance for precision agriculture applications.

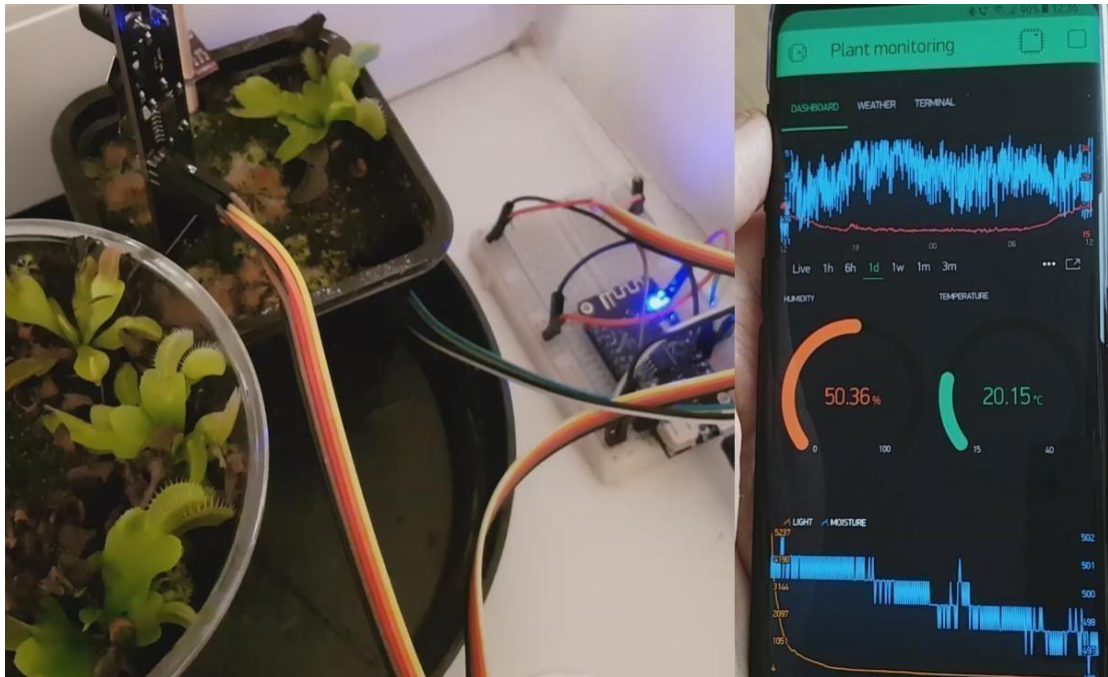


Figure 6.4: Output

CHAPTER – 7

RESULT AND DISCUSSION

7.1 RESULT ANALYSIS

The proposed smart environmental monitoring system using the ESP32 microcontroller was successfully designed and implemented. The system was able to accurately measure key environmental parameters such as temperature, humidity, and soil moisture using appropriate sensors. The data collected from the DHT11/DHT22 and soil moisture sensors was processed by the ESP32 and displayed in real time. The system functioned efficiently, providing continuous monitoring of environmental conditions. The readings obtained were stable and consistent under normal operating conditions.

The integration with IoT platforms such as ThingSpeak and Blynk enabled remote monitoring. Users were able to view real-time data through mobile devices and web applications, making the system highly convenient and user-friendly.

7.2 DISCUSSION

The system demonstrated the effectiveness of using embedded systems and IoT technologies in precision agriculture. By continuously monitoring environmental parameters, the system helps in making informed decisions related to irrigation and crop management. The soil moisture sensor played a key role in identifying the water requirements of the soil, which can help reduce water wastage. Similarly, temperature and humidity data provided insights into environmental conditions affecting plant growth. However, some limitations were observed. The accuracy of the sensors can vary depending on environmental conditions and calibration. The system also depends on a stable internet connection for IoT-based monitoring. Despite these limitations, the system performed reliably and achieved the intended objectives. Overall, the results confirm that the proposed system is efficient, cost-effective, and suitable for real-time environmental monitoring in agricultural applications.

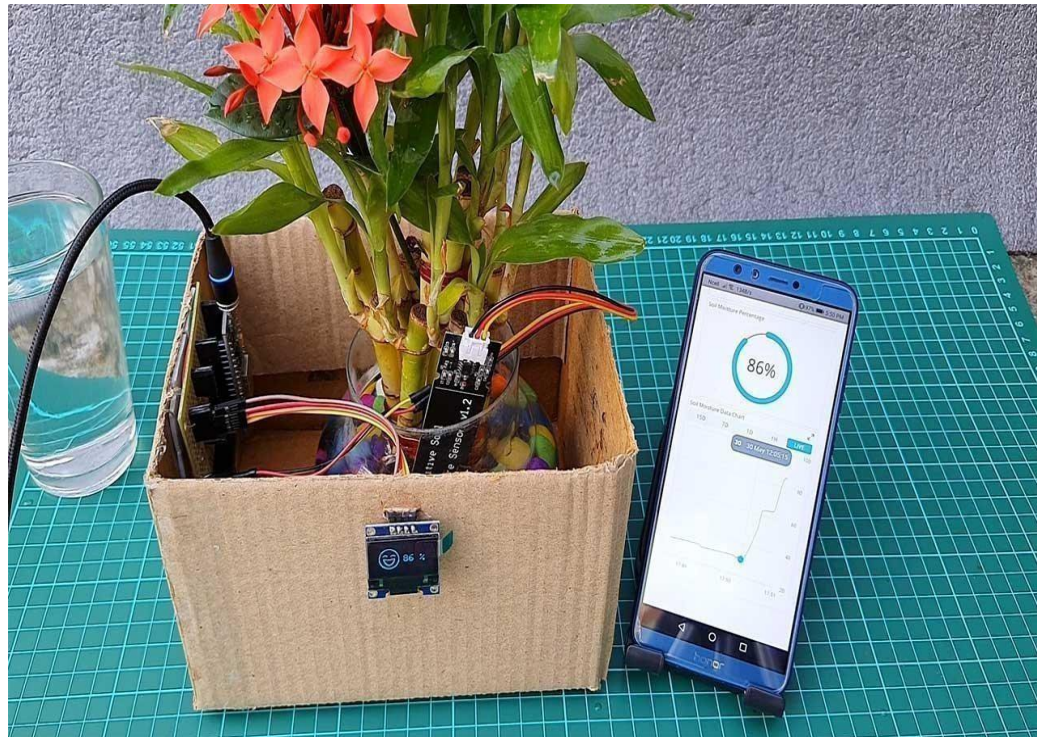


Figure 7.1: Result

ADVANTAGES

1. Real-time monitoring of environmental parameters
2. Reduces manual effort
3. Improves water management and saves resources
4. Remote monitoring through IoT platforms
5. Cost-effective and easy to implement
6. Enhances crop productivity

DISADVANTAGES

1. Depends on internet connectivity
2. Sensor accuracy may vary

3. Requires initial setup and calibration
4. Limited range without network

LIMITATIONS

1. Basic sensors provide moderate accuracy
2. Not fully automated (only monitoring system)
3. Performance may be affected by environmental condition
4. Requires stable power supply
5. Limited scalability without additional components

7.2 COMPARISON: PROPOSED VS EXISTING SYSTEMS

1. System Overview

Aspect	Existing Systems	Proposed System (ESP32-based)
Purpose	Traditional irrigation and monitoring systems; often manual or semi-automated	Real-time, automated monitoring of environmental parameters (humidity, temperature, soil moisture) for precision agriculture
Technology	Wired sensors, PLC-based systems, or low-end IoT solutions	ESP32 microcontroller with Wi-Fi/Bluetooth, IoT-enabled sensors for real-time data collection and transmission
Scope	Limited sensor coverage, delayed data reporting	Continuous monitoring with real-time alerts, remote access, and data logging

2. Data Acquisition & Accuracy

Aspect	Existing Systems	Proposed System
Sensors	Basic sensors, often with low accuracy	High-precision digital sensors integrated with ESP32
Data Frequency	Periodic/manual readings (daily or weekly)	Continuous, real-time data acquisition
Data Quality	Prone to human error	Automated, reducing human error and improving reliability

3. Communication & Connectivity

Aspect	Existing Systems	Proposed System
Connectivity	Often offline; local monitoring only	Wi-Fi/Bluetooth-enabled ESP32 allows cloud integration and remote monitoring
Alerts & Notifications	Manual checks required	Real-time alerts via mobile/web apps

4. Automation & Control

Aspect	Existing Systems	Proposed System
Irrigation Control	Manual or timer-based	Automatic irrigation based on soil moisture thresholds
Decision Support	Farmers rely on experience or periodic data	System provides actionable insights using sensor data trends

5. Power & Cost Efficiency

Aspect	Existing Systems	Proposed System
Power Consumption	High (wired systems, constant operation)	Low-power ESP32 with sleep modes for energy efficiency
Cost	Expensive installation and maintenance	Affordable microcontroller-based setup; scalable and modular

6. Scalability & Integration

Aspect	Existing Systems	Proposed System
Scalability	Adding new sensors is complex and expensive	Easy integration of multiple sensors and expansion to large fields
IoT Integration	Rare	Cloud-based monitoring, data analytics, and mobile app integration possible

7. User Experience

Aspect	Existing Systems	Proposed System
Ease of Use	Requires technical expertise	User-friendly mobile/web dashboards; remote monitoring
Maintenance	Manual and frequent	Automated alerts for maintenance; modular components

CHAPTER – 8

ADVANTAGES AND APPLICATIONS

8.1 ADVANTAGES

1. Real-Time Monitoring

- Continuous measurement of temperature, humidity, and soil moisture.
- Immediate detection of changes for timely action.

2. Remote Access & Control

- Data accessible via smartphone, tablet, or web dashboard.
- Enables remote irrigation control and alerts.

3. Energy Efficient

- Low-power ESP32 microcontroller with sleep modes.
- Compatible with solar power, reducing dependency on grid electricity.

4. Cost-Effective

- Affordable sensors and ESP32 reduce hardware costs.
- Minimizes water and energy wastage through precise irrigation.

5. Automation & Decision Support

- Can trigger automatic irrigation based on soil moisture thresholds.
- Provides actionable insights for crop management and yield optimization.

6. Scalable & Modular

- Easily add more sensors or extend to multiple fields.
- Supports integration with other IoT devices and cloud platforms.

7. Data Logging & Analytics

- Stores historical data for trend analysis.
- Enables predictive farming strategies and improved resource management.

8. Environmental Sustainability

- Reduces water wastage.
- Helps in smart farming practices that lower environmental impact.

8.2 APPLICATIONS

1. Precision Agriculture

- Optimize irrigation schedules.
- Monitor microclimate conditions for crops.

2. Greenhouse Monitoring

- Automate irrigation and climate control.
- Maintain optimal temperature, humidity, and soil moisture.

3. Smart Irrigation Systems

- Trigger irrigation only when soil moisture is below thresholds.
- Integrate with water pumps or drip irrigation systems.

4. Climate & Soil Research

- Collect long-term environmental data for soil health studies.
- Monitor climate conditions in experimental farms.

5. Remote or Off-Grid Farming

- Solar-powered monitoring in areas without reliable electricity.
- Enables smallholders to manage crops efficiently.

6. IoT-Based Farm Management

- Integrates into smart farm networks.
- Supports predictive analytics and AI-based crop management.

CHAPTER – 9

CONCLUSION AND FUTURE SCOPE

9.1 CONCLUSION

The proposed ESP32-based smart environmental monitoring system provides a significant improvement over traditional systems in precision agriculture. By enabling real-time monitoring of humidity, temperature, and soil moisture, it allows farmers to make informed decisions, optimize irrigation schedules, and reduce resource wastage. The system's remote access, automation, and low-power operation make it suitable for modern farming needs, improving both productivity and sustainability.

In summary, the system:

- Enhances crop yield and quality through precise environmental control.
- Reduces manual labor and operational costs by automating data collection and irrigation.
- Supports eco-friendly farming practices by conserving water and energy.
- Provides a scalable and modular platform for integration with future IoT solutions. The project demonstrates how IoT and microcontroller technology (ESP32) can transform agriculture into a smarter, more data-driven practice.

9.2 FUTURE SCOPE

1. Integration with Advanced IoT Platforms

- Connect multiple ESP32 units to a cloud-based dashboard for centralized farm management.
- Enable AI-based predictive analytics for crop yield optimization.

2. Additional Sensor Integration

- Include Ph sensors, light intensity sensors, and nutrient sensors for comprehensive soil and crop monitoring.
- Monitor pest infestation or disease indicators using imaging sensors.

3. Automation Enhancements

- Integrate with drip irrigation, fertigation, or greenhouse climate control systems.
- Enable fully autonomous irrigation based on machine learning algorithms.

4. Energy Optimization

- Combine solar and battery backup systems for remote or off-grid farming.
- Implement energy-efficient data transmission protocols (LoRa, NB-IoT).

5. Mobile and Web Applications

- Develop advanced apps for real-time alerts, historical data analysis, and predictive recommendations.
- Allow multiple user roles for farm managers and agricultural consultants.

6. Scalable Smart Agriculture Networks

- Expand to large-scale farms or smart cities with multiple farms connected via IoT networks.
- Enable real-time regional agricultural monitoring for government or cooperative initiative.

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